Capstone Progress Update 2: Weeks 3 and 4

An Automated Data Pipeline for the German Electricity Market

1. Where I stand

In these two weeks I cleaned the data, added quality checks, and answered the first three of my five questions. The history now runs from October 2018 to today, which is 70,729 hours of day-ahead prices for Germany and Luxembourg. All 19 data-quality checks pass. I am on track.

2. What I did

Week	What I did	How it went
3	Built the clean tables on an hourly time step, in Berlin local time	Prices, generation, load and temperature now line up hour by hour
3	Added quality checks for gaps, duplicates and impossible values	19 checks, and all of them pass
3	Built the analysis tables and the monitoring table	Ready for the questions and for the reliability check
4	Answered question 1: prices by hour, weekday and season	Clear daily and yearly patterns, see below
4	Answered question 2: how renewable share moves the price	A stable and strong relationship, see below
4	Answered question 3: the best time for flexible electricity use	Cheap and green hours mostly coincide, see below

3. Question 1: how do prices change by hour, weekday and season?

I looked at the last 12 months for the daily pattern, and at all years for the seasons.

- **Hour of day.** The cheapest hour is 13:00 at 50.8 EUR/MWh on average, the most expensive is 19:00 at 160.9 EUR/MWh. The midday dip comes from solar power. The evening peak comes when solar fades while demand is still high.
- **Weekday.** Weekdays average 117.8 EUR/MWh and weekends 85.9 EUR/MWh, which is 27% lower, because industry and offices use less power.
- **Season.** Mean prices are 93.9 in winter, 79.5 in spring, 104.2 in summer and 102.8 in autumn (EUR/MWh). Negative prices happen mostly in spring and summer: 5.5% of spring hours and 4.7% of summer hours, against 2.3% in winter.
- **Over the years.** The share of negative-price hours grew from 2.4% in 2019 to 6.6% in 2025. The average daily spread between the highest and lowest price grew from 30 to 124 EUR/MWh. The yearly average peaked in 2022 at 235 EUR/MWh because of the gas crisis, so results that pool all years are pulled up by 2021 to 2023. That is why I use the last 12 months as the better guide to today's pattern.

4. Question 2: how strongly does renewable share move the price?

I used a regression of the hourly price on the renewable share of load, load and temperature. It also accounts for hour of day, weekday, holidays and the month, which takes out slow changes in fuel and CO_2 prices that are not in my data. Standard errors allow for the fact that neighbouring hours are related (Newey-West, 24 lags).

Sample	Effect of +10 points renewable share	95% interval	R^2	Hours used
Since January 2024	-15.9 EUR/MWh	-16.5 to -15.3	0.76	24,673
Whole period	-16.4 EUR/MWh	-17.0 to -15.7	0.82	70,706

- An hour with a 10-point higher renewable share has a price about 15.9 EUR/MWh lower. The average price in that period was 92 EUR/MWh, so this is about 17% of the average. The two estimates are statistically indistinguishable, so the relationship has been stable.
- Each extra GW of load goes with +1.9 EUR/MWh, and each extra degree Celsius with +0.26 EUR/MWh. In a second model, each GW of residual load (load not covered by wind and solar) goes with +3.1 EUR/MWh (3.0 to 3.2). That tells the same story from the other side.
- National holidays were not clearly different from other days once everything else is held fixed.

How to read this. These are associations, not proof of cause. The mechanism is plausible: wind and solar have almost no running cost and push expensive plants out of the market. But the model leaves out cross-border flows, fuel prices within a month, plant outages and the forecasts traders use. The temperature effect is entered as a straight line, although the true effect is U-shaped, because of heating and cooling.

5. Question 3: when is the best time to run flexible electricity use?

Over the last 12 months, the cheapest hours are 13:00, 14:00 and 12:00, and the greenest hours are 13:00, 12:00 and 14:00. The most expensive are 19:00, 20:00 and 21:00. Across the 24 hours, mean price and mean renewable share are strongly negatively correlated (r = -0.91). In today's German market, cheap and green mostly coincide, because solar power drives both.

To see what this is worth, I simulated a load of 1 MW for 3 hours that normally starts at 18:00 and may start up to 6 hours earlier or later. Each day the start is chosen from the day-ahead prices, which are published the day before, so this could be done in practice.

Scenario (364 days)	Cost per run	Renewable share of the run
Usual start at 18:00	466 EUR	47%
Cheapest start	138 EUR	not reported
Greenest start	145 EUR	90%

Moving to the cheapest start saves 328 EUR per run (70%), or 119,319 EUR over the 364 days. It paid off on every day, and the most frequent best shift was 6 hours earlier. The saving is largest for loads that normally start at 18:00.

Limits. The savings use past prices and assume perfect knowledge of the next day's auction result, so they are not a promise about the future. The scenario ignores network charges and taxes, which make up most of a retail bill, and it assumes the load can really be moved.